

Ejemplo $A = QR$ con Householder

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 1 \\ -2 & 0 & -1 \end{bmatrix}$$

Paso 1:

- $a_1 = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$
- $\|a_1\| = \sqrt{1+4+4} = \sqrt{9} = 3$
- $u_1 = a_1 + \|a_1\|e_1 = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ -2 \end{bmatrix}$
- $\rho = \frac{2}{u_1^T u_1} = \frac{2}{\begin{bmatrix} 4 & 2 & -2 \end{bmatrix}^T \begin{bmatrix} 4 \\ 2 \\ -2 \end{bmatrix}} = \frac{2}{16+4+4} = \frac{2}{24} = \frac{1}{12}$
- $u_1 u_1^T = \begin{bmatrix} 4 \\ 2 \\ -2 \end{bmatrix} \begin{bmatrix} 4 & 2 & -2 \end{bmatrix}^T = \begin{bmatrix} 16 & 8 & -8 \\ 8 & 4 & -4 \\ -8 & -4 & 4 \end{bmatrix}$
- $H_1 = I_3 - \frac{2u_1 u_1^T}{u_1^T u_1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{12} \begin{bmatrix} 16 & 8 & -8 \\ 8 & 4 & -4 \\ -8 & -4 & 4 \end{bmatrix} = \begin{bmatrix} 1 - \frac{4}{3} & -\frac{2}{3} & \frac{2}{3} \\ -\frac{2}{3} & 1 - \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & 1 - \frac{1}{3} \end{bmatrix}$

$$H_1 = \frac{1}{3} \begin{bmatrix} -1 & -2 & 2 \\ -2 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$
- $A_1 = H_1 A = \frac{1}{3} \begin{bmatrix} -1 & -2 & 2 \\ -2 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 1 \\ -2 & 0 & -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} -1-4-4 & -3-6 & -5-2-2 \\ -2+4-2 & -6+6 & -10+2-1 \\ 2+2-4 & 6+3 & 10+1-2 \end{bmatrix}$

$$A_1 = \begin{bmatrix} -3 & -3 & -3 \\ 0 & 0 & -3 \\ 0 & 3 & 3 \end{bmatrix}$$

Paso 2:

- $\hat{A}_1 = \begin{bmatrix} 0 & -3 \\ 3 & 3 \end{bmatrix}$

- $\hat{a}_1 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

- $\|\hat{a}_1\| = \sqrt{0+9} = \sqrt{9} = 3$

- $u_2 = \hat{a}_1 + \|\hat{a}_1\|e_1 = \begin{bmatrix} 0 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$

- $\rho = \frac{2}{u_2^T u_2} = \frac{2}{\begin{bmatrix} 3 & 3 \end{bmatrix}^T \begin{bmatrix} 3 \\ 3 \end{bmatrix}} = \frac{2}{9+9} = \frac{2}{18} = \frac{1}{9}$

- $u_2 u_2^T = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \begin{bmatrix} 3 & 3 \end{bmatrix}^T = \begin{bmatrix} 9 & 9 \\ 9 & 9 \end{bmatrix}$

- $\hat{H}_2 = I_2 - \frac{2u_2 u_2^T}{u_2^T u_2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 9 & 9 \\ 9 & 9 \end{bmatrix} = \begin{bmatrix} 1-1 & 0-1 \\ 0-1 & 1-1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

- $H_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

- $A_2 = H_2 A_1 = \frac{1}{3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} -3 & -3 & -3 \\ 0 & 0 & -3 \\ 0 & 3 & 3 \end{bmatrix}$

$$A_2 = \begin{bmatrix} -3 & -3 & -3 \\ 0 & -3 & -3 \\ 0 & 0 & 3 \end{bmatrix}$$

$$R = A_2 \Rightarrow R = \begin{bmatrix} -3 & -3 & -3 \\ 0 & -3 & -3 \\ 0 & 0 & 3 \end{bmatrix}$$

$$H_1 H_2 = Q \Rightarrow Q = \frac{1}{3} \begin{bmatrix} -1 & -2 & 2 \\ -2 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \Rightarrow Q = \frac{1}{3} \begin{bmatrix} -1 & -2 & 2 \\ -2 & -1 & -2 \\ 2 & -2 & -1 \end{bmatrix}$$

Comprobación $A = QR$

$$QR = \frac{1}{3} \begin{bmatrix} -1 & -2 & 2 \\ -2 & -1 & -2 \\ 2 & -2 & -1 \end{bmatrix} \begin{bmatrix} -3 & -3 & -3 \\ 0 & -3 & -3 \\ 0 & 0 & 3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 & 3+6 & 3+6+6 \\ 6 & 6+3 & 6+3-6 \\ -6 & -6+6 & -6+6-3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 & 9 & 15 \\ 6 & 9 & 3 \\ -6 & 0 & -3 \end{bmatrix}$$

$$QR = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 1 \\ -2 & 0 & -1 \end{bmatrix} = A$$